



Analyze *E-Commerce*

Sales to Improve Customer Conversion &
Retention

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INTRODUCTION :

This project analyzes e-commerce sales data to understand customer purchasing behavior and improve customer conversion and retention. Using **Python, SQL, Power BI, and Machine Learning**, the project identifies repeat buying patterns, segments customers, and provides actionable business insights to support data-driven decision-making.

PROBLEM STATEMENT



Business Objectives

Analyze Sales
Performance

Analyze
Customer
Retention

Optimize
Pricing &
Operations

*Improve
Customer
Satisfaction*

Business Questions

- Which product categories generate the highest sales revenue and order volume?
- Can a sample of customer orders reliably represent the overall sales performance?
- Is there a significant difference in the Average Final Price between New and Loyal customers?
- Can we predict whether a customer will become a Repeat Buyer? (Logistic Regression)
- Can similar customer purchase patterns be used to classify Repeat Buyers? (KNN)
- Can customers be grouped into distinct segments based on their purchasing behaviour?
- Which business variables have the strongest relationship with Final Price?
- How does Unit Price influence the Final Price? (Simple Linear Regression)
- How do Unit Price, Quantity, Discount (%), and Shipping Cost collectively influence the Final Price? (Multiple Linear Regression)
- What is the probability that an order is delayed and receives Low Customer Satisfaction? (Probability)
- What is the probability of High Customer Satisfaction given On-Time Delivery? (Conditional Probability)
- Is there a significant association between Delivery Status and Customer Satisfaction? (Chi-Square Test)



RetailTech E-Commerce Sales Dataset



Description

- Tables used are Customer, order, product, review, and location data
- Used to analyze customer conversion & retention

DATASET OVERVIEW

Dataset Summary

- **Records:** 1,847,673
- **Features:** 49
- **Target Variable:** Repeat Buyer
- **Merged from 6 datasets**



Tables Used

- Customers
- Orders
- Order Items
- Products
- Reviews
- Geolocation



DATA PRE-PROCESSING (python)



01

Checked & treated missing values.

02

Converted data into appropriate data types

03

Detected & handled outliers

04

Encoded categorical variables

05

Standardized numerical features

06

Prepared clean data for EDA, Power BI & ML

07

Merged 6 datasets into a Master Dataset

08

Connected Python to MySQL

SQL ANALYSIS

To automate business reporting and improve database efficiency using **SQL Views, Stored Procedures, and Triggers.**

VIEWS

Category-wise Sales Performance (BO1)

- Displays category-wise sales for business reporting.

Customer Retention View (BO2)

- Monitors repeat customer behaviour.

Customer Satisfaction View (BO4)

- Shows customer satisfaction across product categories.

STORED PROCEDURES

Display Category-wise Sales Performance (BO1)

- Generates category-wise sales reports automatically.

Customer Purchase History (BO2)

- Displays complete customer purchase history.

Month-wise Sales Summary (BO3)

- Generates monthly sales summary reports.

TRIGGERS

Delivery Performance Trigger (BO1)

- Automatically classifies orders as On-Time or Delayed.

Repeat Buyer Trigger (BO2)

- Automatically updates the Repeat Buyer status.

Customer Satisfaction Trigger (BO4)

- Automatically assigns High or Low Satisfaction based on review score.

EXCEL DASHBOARD



RetailTech Analytics
Customer Conversion & Retention
Dashboard

E-Commerce Sales Analysis for Customer Conversion & Retention

Total Revenue
11794729452

Total Orders
138017

Average Order Value
12119.21081

Total Customers
134407

Repeat Buyer Rate
46 %

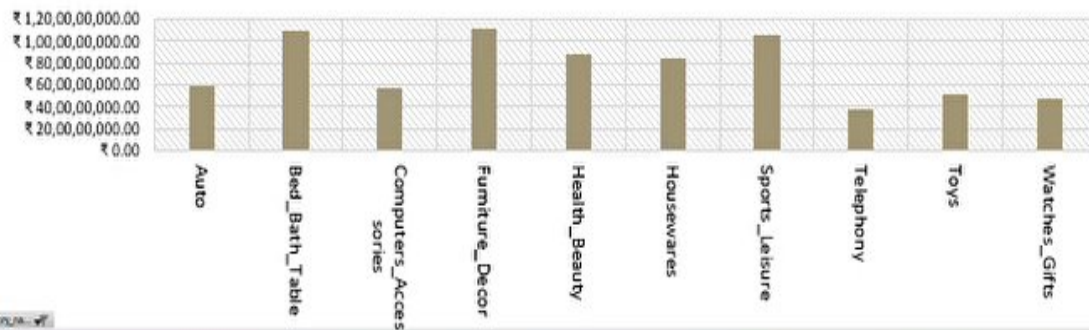
REPEAT BUYER DISTRIBUTION BY CUSTOMER SEGMENT



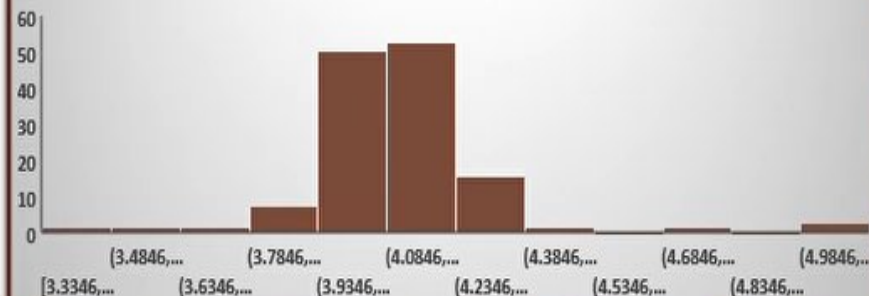
Customer Segment



Revenue by Category



Total reviews



Dashboard Insights

- New customers (50%) form the largest customer segment.
- Bed_Bath_Table is the highest revenue-generating category.
- Customer reviews are mostly 4-star, reflecting positive satisfaction.

Dashboard

Customer Satisfaction

Customer Retention

Sales Summary

Ecommerce Sales Analysis Dashboard

E-Commerce Sales Analysis for Customer Conversion & Retention

Sales Performance

11.79bn

Total Revenue

65K

Total Orders

182.64K

Average Order V...

10.45bn

Total Estimated Pro...

Customer Retention

41K

Total Customers

203.01

Average Days Betw...

3.35

Average Purchas...

20K

Repeat Customers

Profitability

161.89K

Average Profit per Order

13.50

Average Discount %

0.89

Profit Margin %

10.45bn

Total Estimated Profit

Customer Satisfaction

4.09

Average of review score

77.10

Positive Review Rat...

25.93

On-Time Delivery R...

14.43

Average Delivery...

Sales Performance Dashboard

Sales Performance by Category

Category_name	Total Orders	Total Revenue	Average Order Value
Watches_Gifts	1274	10,24,10,044.12	80,384.65
Toys	1483	10,89,38,884.08	73,458.45
Telephony	1104	8,14,74,931.86	73,799.76
Tablets_Printing_Image	14	13,56,324.78	96,880.34
Stationery	1587	4,91,80,536.43	30,989.63
Sports_Leisure	2683	21,27,56,652.89	79,298.04
Small_Appliances_Home_Oven_And_Coffee	6	25,50,609.07	4,25,101.51
Small_Appliances	261	1,86,59,854.91	71,493.70
Signaling_And_Security	64	62,72,514.32	98,008.04
Total	13790	2,53,29,73,074.14	1,83,681.88

Sales performance is driven by a few high-performing product categories, while most orders are successfully delivered, indicating strong operational efficiency.

Category_name

All

Year

2021

2022

2023

2024

2025

2026

Question : OBJ1

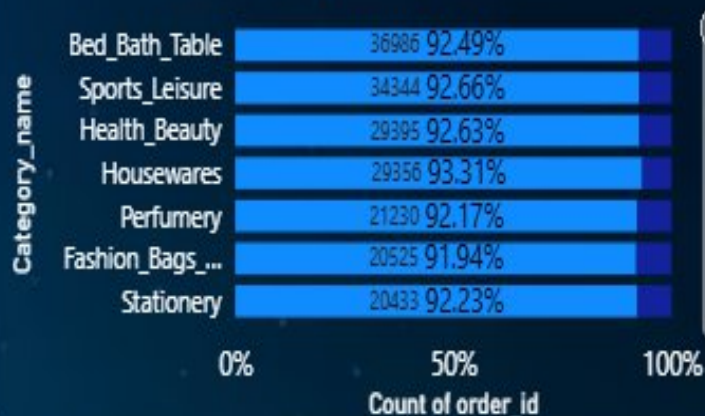
Which product categories contribute the highest sales revenue and order volume, and how does order status vary across categories?

INTERPRETATION:

- Telephony generated the highest revenue of ₹814.75 million.
- Sports & Leisure recorded the highest order volume with 2,683 orders.
- 84,971 orders were successfully delivered, indicating strong sales performance and efficient order fulfilment.

Count of order_id by Category_name and order_status

order_status ● Delivered ● Returned



Total Revenue by Category_name



Customer Retention

Count of customer_id by customer_segment



Customer retention analysis shows that repeat buyers slightly outnumber one-time buyers, with a few product categories driving most repeat purchases and strengthening long-term customer loyalty.

Customer by Category and Repeat Buyer



Impact of Delivery Status on Repeat Buyers



BO2 – Analyze Customer Business Question:

How do customer segments and delivery performance influence repeat purchases and long-term customer retention?

Interpretation

- **New customers** form the largest customer segment.
- **Loyal customers** show the shortest average time between purchases.
- **On-time deliveries** contribute to higher repeat purchases and stronger retention.



Profitability

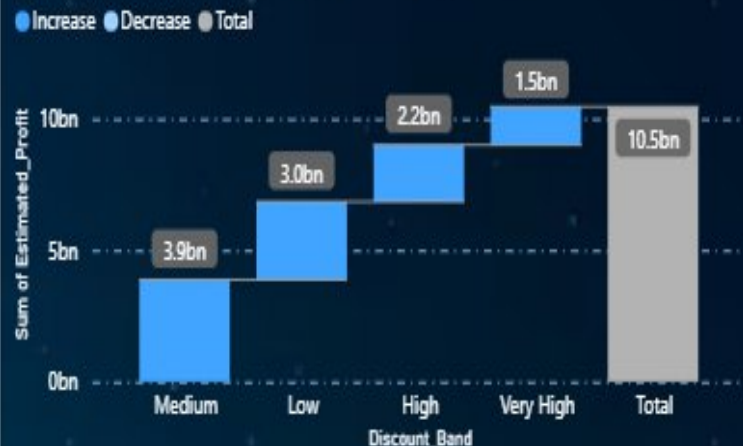


Average of Final_Price and Average of Estimated_Profit by Discount_Band

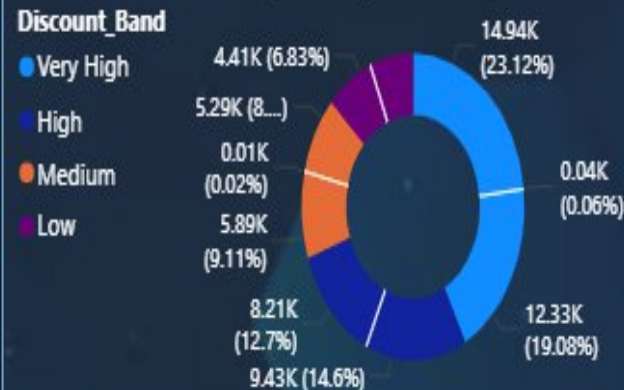


Profitability analysis indicates that discount strategies significantly influence both sales and estimated profit, helping identify the optimal discount levels for maximizing

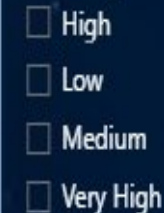
Sum of Estimated_Profit by Discount_Band



Average of Final_Price, Average Discount % and Average of Estimated_Profit by Discount_Band



Discount_Band



BO3:Optimize Pricing & Operations

Business Question:

How do different discount levels impact Final Price and Estimated Profit?

Interpretation

- Very High discounts generated the highest Final Price and Estimated Profit.
- Estimated Profit varies across different discount bands.
- The right discount strategy can improve both sales and profitability.

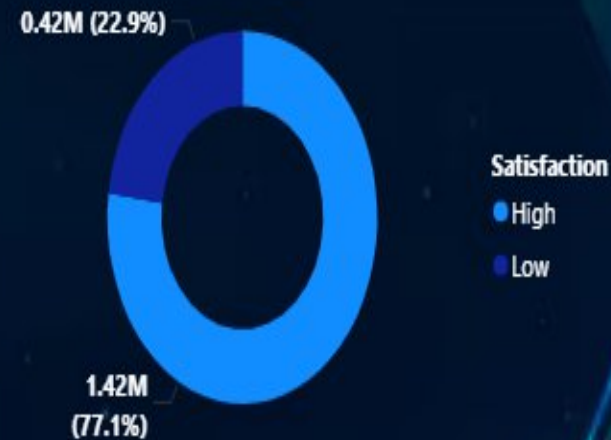
Customer Satisfaction

Average of Review Score

Satisfaction × review_id × Repeat_Buyer ×
Low R00002



Count of review_score by Satisfaction



Average of review_score by Delivery_Status



Customer satisfaction remains positive, with higher review scores reflecting a satisfactory shopping experience and opportunities to

Year

- ☐ 2021
- ☐ 2022
- ☐ 2023
- ☐ 2024
- ☐ 2025
- ☐ 2026

BO4:Improve Customer Satisfaction

Business Question:

How do review scores and delivery performance influence overall customer satisfaction?

Interpretation

- 77.1% of customers reported **high satisfaction**.
- The average review score remained around **4.0**, indicating a positive customer experience.
- Customer satisfaction is consistently high across different delivery statuses.

OBJ4 – Improve Customer Satisfaction: **PROBABILITY ANALYSIS**

Joint Probability

Question

What is the probability that an order is delayed and receives low customer satisfaction?

Result

P= 0.1712

Interpretation

There is a **17.12%** probability **both delayed and receives a low customer satisfaction rating**, indicating a direct association

Business Insight

Reducing delivery delays can help **decrease customer dissatisfaction**, improve service quality, and strengthen customer retention.

Conditional Probability

Question

What is the probability that a customer is highly satisfied given the order was delivered on time?

Result

P= 0.7769

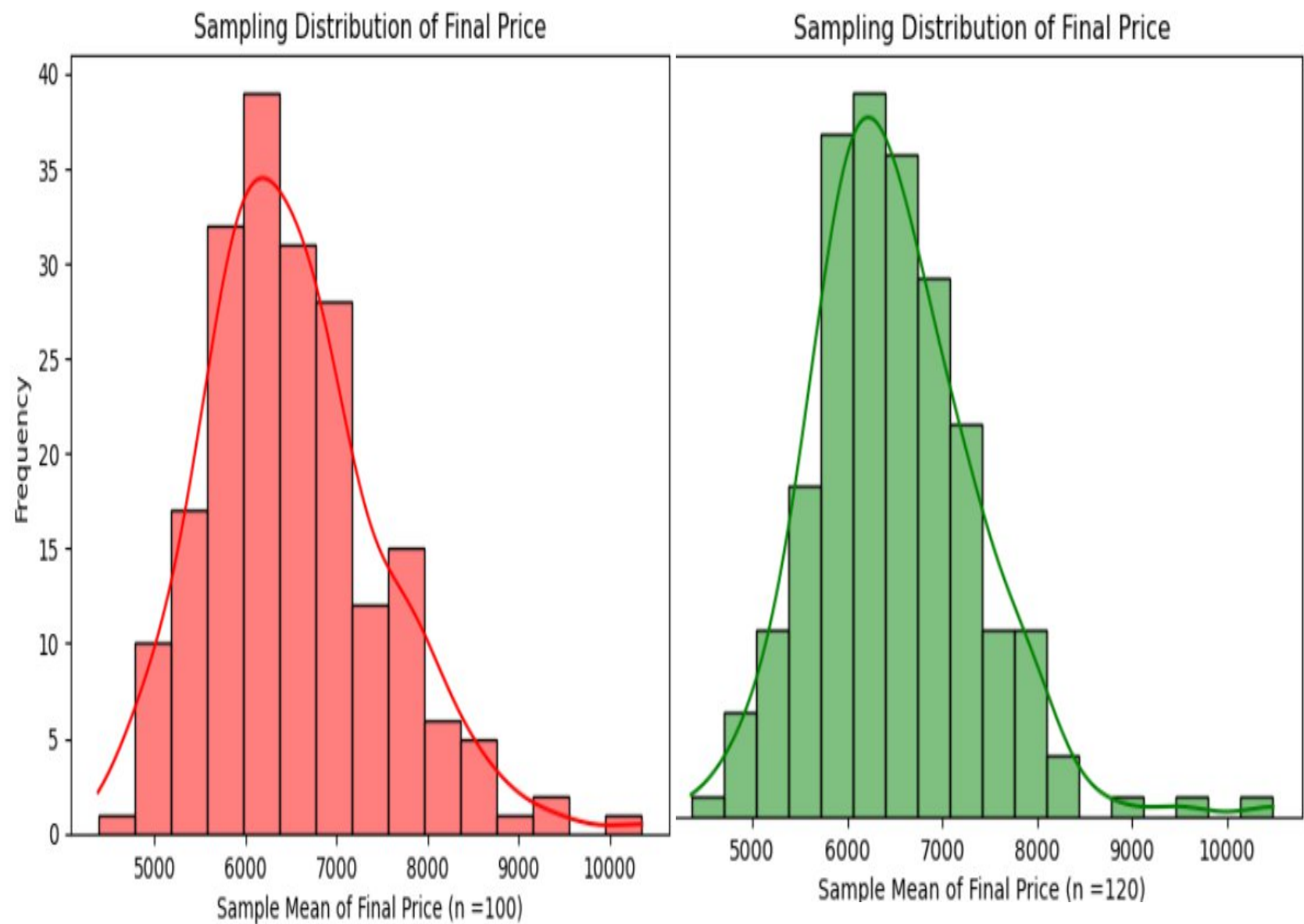
Interpretation

There is a **77.69%** probability that customers provide **high satisfaction ratings when their orders are delivered on time**

Business Insight

Improving on-time delivery performance can significantly **increase customer satisfaction**, encourage repeat purchases, and enhance customer loyalty.

Central Limit Theorem (CLT)



Metric	Value
Population Mean	6383.57
Mean of Sample Means	6522.27
Population Standard Deviation	5068.09
Standard Deviation of Sample Means	1192.96

Interpretation

The sample mean is close to the population mean, and the lower sampling variability confirms that the sample reliably represents the population, validating the Central Limit Theorem.

OBJECTIVE 2: Analyze Customer Retention

TWO-SAMPLE Z-Test

QUESTION:

Is there a significant difference in the Average Final Price between New and Loyal customers?

H0: There is **no significant difference**

H1: There **is a significant difference** in the Average Final.

Interpretation

New customers spend slightly more per order than Loyal customers, indicating that customer type has a measurable impact on average order value.

Decision: ✓ **Reject H_0**

Result: A **statistically significant difference exists** between the Final Price of New and Loyal customers.

Business Recommendation

- Provide personalized offers to convert New customers into repeat buyers.
- Introduce exclusive loyalty rewards to increase repeat purchases and customer retention.

Business Objective 4 – Improve Customer Satisfaction

CHI-SQUARE Test of Independence

Business Question

Is there a significant association between Delivery Status and Customer Satisfaction (Review Score)?

H_0 : Delivery Status and Review Score are independent

H_1 : Delivery Status and Review Score are not independent

Metric	Value
Test	Chi-Square Test
Significance Level (α)	0.05
Decision	Reject H_0

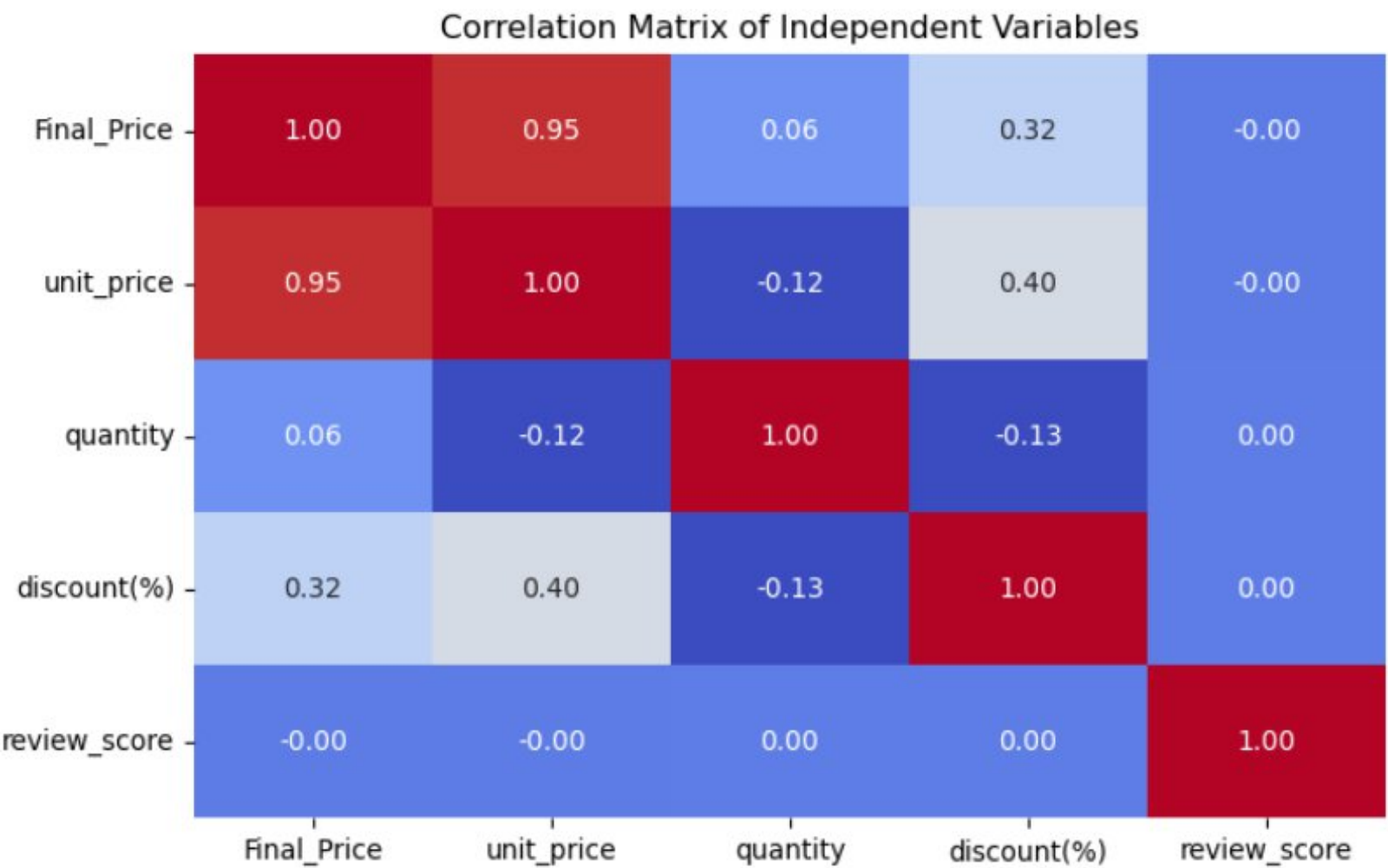
Interpretation

Since $p\text{-value} < 0.05$, The Chi-Square Test shows a significant relationship between Delivery Status and Customer Satisfaction. On-time deliveries are associated with higher customer satisfaction

OBJECTIVE 3 – Optimize Pricing & Operation: CORRELATION ANALYSIS

Business Question

Which independent variables are most suitable for building the regression models to predict Final Price?



Analysis

Variable	Correlation with Final Price
Unit Price	0.95 (Strong Positive)
Discount (%)	0.32 (Moderate Positive)
Quantity	Weak Correlation
Review Score	Negligible Correlation

Interpretation :

The selected independent variables exhibit low multicollinearity, making them suitable for regression modelling.

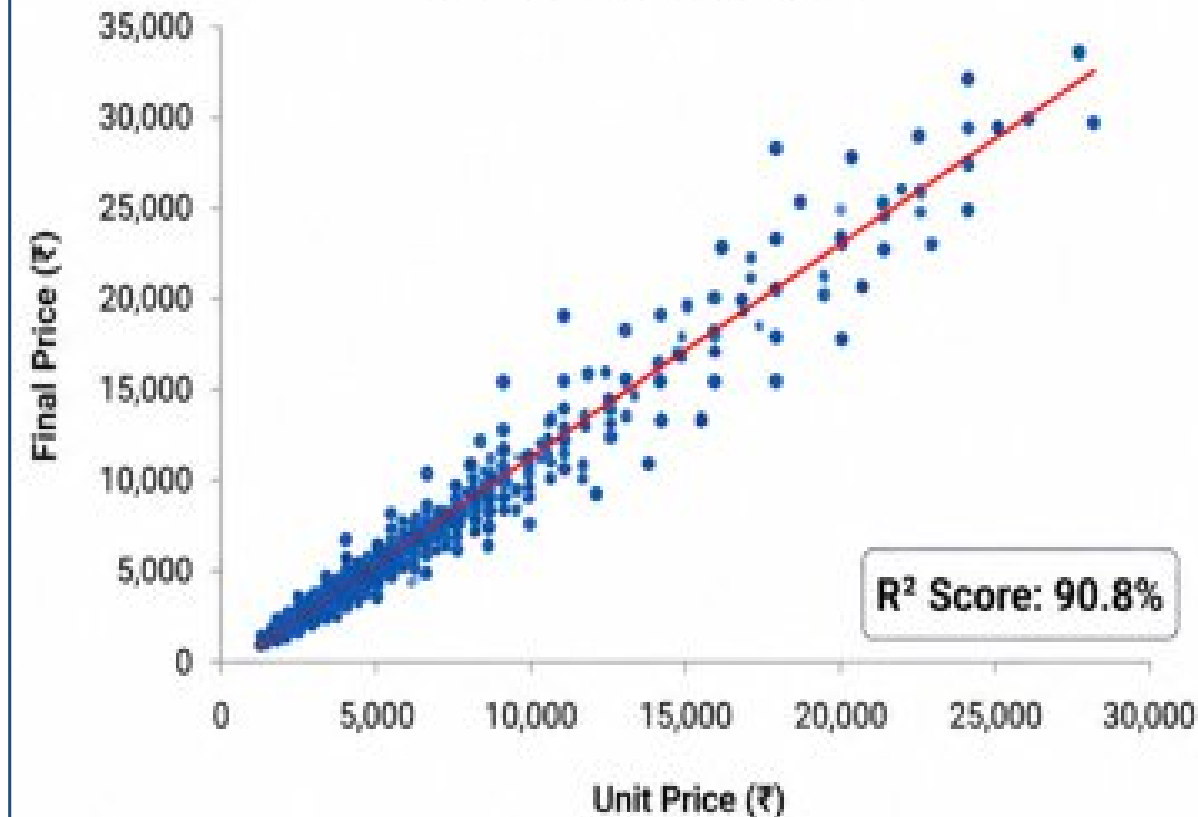
OBJECTIVE 3 – Optimize Pricing & Operations : SIMPLE LINEAR REGRESSION

QUESTION:

How accurately can Unit Price predict the Final Price of future orders?

MODEL VISUALIZATION

Unit Price vs Final Price



Key Results

- **Independent Variable:** Unit Price
- **Dependent Variable:** Final Price
- **Model:** Simple Linear Regression
- **R² Score:** 90.8%

Interpretation

- **Unit Price** is the strongest predictor of **Final Price**.
- As **Unit Price** increases, the **Final Price** also increases proportionally.
- The high **R² value** indicates that the model provides an excellent fit to the data.

Business Insights

- **Unit Price** is the primary driver of **Final Price**.
- Businesses can use this model to estimate product prices accurately.
- Optimising pricing strategies can improve revenue forecasting and pricing decisions.

BO3 :Optimize Pricing & Operations : MULTIPLE LINEAR REGRESSION

QUESTION

How accurately can the Final Price of future orders be predicted using Unit Price, Quantity, Discount (%), and Shipping Cost?

Model Performance

Metric	Value
Testing R ²	94.40%
MAE	1109.21
RMSE	2175.19
MSE	4,731,443.86

Business Recommendation

- Use multiple product and customer attributes for accurate sales forecasting.
- Optimize pricing, discounts, and shipping strategies to maximize revenue.
- Apply the model to support pricing decisions, demand planning, and revenue forecasting

- **Dependent Variable (Y):** Final Price
- **Independent Variables (X):**
 - Unit Price
 - Quantity
 - Discount (%)
 - Shipping Cost
- **Model Accuracy (R²): 94.4%**
- **Model Performance:** Higher predictive accuracy than Simple Linear Regression.

Interpretation

- Unit Price remains the strongest predictor, while Quantity, Discount (%), and Shipping Cost further improve the prediction accuracy.
- Considering multiple variables provides a more reliable and realistic pricing model than using a single predictor.

OBJ 2 – Analyze Customer Retention: Logistic Regression &

Question:-Can we predict whether a customer will become a Repeat Buyer?

Logistic Regression

Metric	Value
Accuracy	73.97%
Precision	80.67%
Recall	73.70%
F1 Score	77.03%
ROC-AUC	81.38%

Interpretation : Logistic Regression

- Achieved **73.97% accuracy** in predicting Repeat Buyers.
- **ROC-AUC of 81.38%** indicates strong discrimination between Repeat and Non-Repeat Buyers.
- The model is easy to interpret and identifies the key factors influencing customer retention.

QUESTION:
Can customer purchasing behaviour be used to classify customers as Repeat Buyers or Non-Repeat Buyers?

K-Nearest Neighbors (KNN)

Metric	Value
Accuracy	69.11%
Precision	83.81%
Recall	78.26%
F1 Score	80.94%
Best K	9

Interpretation: KNN

- Achieved **69.11% accuracy** with **Best K = 9**.
- Higher **Precision (83.81%)** and **F1 Score (80.94%)** indicate effective classification performance.
- Predicts Repeat Buyers based on similarity in customer purchasing behavior.

BO2 – Analyze Customer Retention K-MEANS CLUSTERING



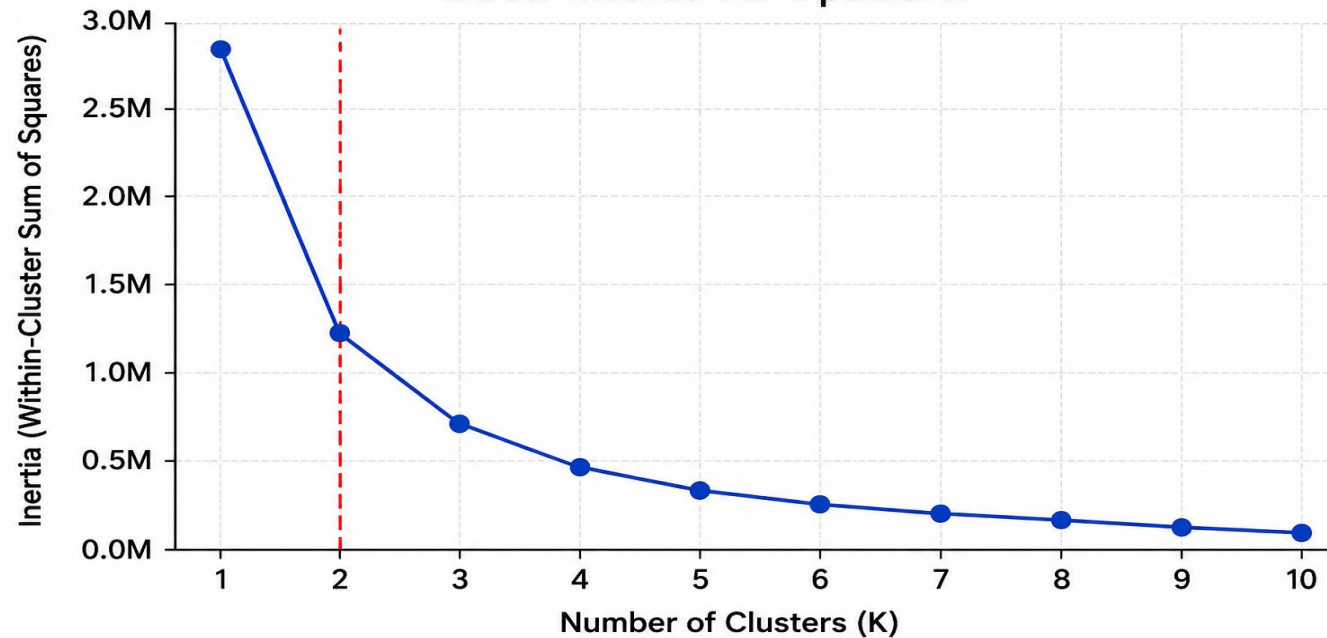
Business Question

Can customers be grouped into distinct segments based on their purchasing behaviour to improve retention strategies?

Optimal Number of Clusters

Elbow Method: $K = 2$

Elbow Method For Optimal K



Business Recommendation:

Retain loyal customers, encourage bulk purchases, reward premium customers, and improve service for dissatisfied customers.

Cluster Distribution

Cluster	Customers	Interpretation
Cluster 0	1,816,792	High-value customers
Cluster 1	30,881	Regular customers

Metric	Result
Silhouette Score	0.723
Davies-Bouldin Index (DBI)	0.584
Calinski-Harabasz Index (CHI)	758,422.44

BUSINESS INSIGHTS



Pricing Drives Revenue

- Unit Price is the strongest factor influencing Final Price.
- It explains **90.8% ($R^2 = 90.8\%$)** of the variation in Final Price.



Accurate Sales Prediction

- Multiple Linear Regression achieved **94.4% prediction accuracy**.
- Multiple business variables improved forecasting reliability.



Customer Retention

- Logistic Regression and KNN successfully predicted **Repeat Buyers**.
- These models help identify customers for targeted retention campaigns.



Delivery Impacts Satisfaction

- Customers receiving **on-time deliveries** have a **77.69% probability** of being satisfied. .



Customer Segmentation

- K-Means Clustering identified **2 distinct customer segments**.
- **Premium Customers (Cluster 1)** contribute the highest revenue and should be prioritised through loyalty programmes and personalised offers.



PROJECT SUMMARY



01 Integrated Multiple Datasets

- Merged multiple e-commerce datasets into a single **Master Dataset**.
- Established relationships for comprehensive business analysis.

02 Data Cleaning & Preparation

- Cleaned missing values, removed duplicates, and handled data inconsistencies.
- Performed exploratory data analysis (EDA) in python.

03 Dashboard & SQL Implementation

- Developed an interactive **Excel Dashboard** for KPI reporting.
- Created **SQL Views, Stored Procedures, and Triggers** for efficient data management.

04 Applied Advanced Analytics

- Applied **Probability, Hypothesis Testing, Regression, Classification, and Clustering** techniques.
- Built predictive models to analyze customer behaviour and business performance.

05 Generated Actionable Insights

- Identified key factors affecting customer conversion and retention.
- Provided data-driven recommendations to improve pricing, marketing, and operational decisions.

FUTURE SCOPE

This project can be enhanced by integrating **real-time e-commerce data** for live business insights and dashboards. Advanced machine learning models such as **Random Forest, XGBoost, and Gradient Boosting** can improve prediction accuracy. Incorporating **seasonal trends, promotional campaigns, and competitor pricing** can enable better forecasting. Additionally, developing a **personalised recommendation system** and deploying the solution as a **web-based BI platform** can support real-time, data-driven business decisions.

LIMITATIONS

- Based on historical e-commerce data and may not reflect current market trends.
- Missing values and data quality issues may affect analysis accuracy.
- External factors such as seasonality, promotions, and competitor pricing were not considered.
- Machine learning models require periodic retraining with new data to maintain prediction accuracy.

CONCLUSION

- **Sales Performance:** Identified top-performing product categories and key factors influencing sales and customer purchasing behaviour.
- **Customer Retention:** Analysed repeat purchase behaviour and identified **high-value, regular, and at-risk customer segments** using K-Means clustering.
- **Pricing & Operations:** Found that **Unit Price is the strongest predictor of Final Price**, while discounts and shipping costs influence pricing and operational performance.
- **Customer Satisfaction:** Established a **significant relationship between delivery performance and customer satisfaction**, showing the importance of timely delivery.
- **Customer Prediction:** Logistic Regression and KNN successfully classified customers as **Repeat Buyers and Non-Repeat Buyers**, supporting targeted retention strategies.



Thank you so much For watching